Chapter 162

Resolve FX Refine

These plugins let you make different kinds of targeted improvements.

Contents

COLOR

Beauty (Studio Version Only)

Beauty is a plugin that lets you control texture. In Ultra Beauty mode, you can selectively smooth image detail that falls above a particular threshold, while preserving detail falling below a particular threshold. In this way, you can smooth larger textures you don't want, while either preserving or exaggerating smaller textures that you do want.

In the case of faces and skin, this plugin provides another method of smoothing larger, unwanted blemishes, while preserving desirable edge detail and underlying structures such as pores, so you can improve the complexion of subjects on shoots where hair and makeup was not available, without over-softening realistic skin details and creating an overly fake plastic look.

(Left) The original image, (Right) A face with Beauty applied to soften the complexion while retaining fine facial detail

Please note that this plugin is not only for skin detail, and it's not only for softening. Once you isolate the fine detail you want to preserve in a subject, you also have the option of exaggerating it to create highly textured results. This plugin is effective for any subject with textures that need refinement.

> **TIP:** This plugin is best used within a qualification that isolates the particular feature you're trying to smooth. The default settings apply a moderate amount of smoothing and fine detail recovery that's appropriate for faces in closeup but will not be universally appropriate for all situations, so some fine-tuning will always be necessary. For naturalistic results, this plugin produces the best results when used in moderation.

Operating Mode

The Beauty plugin has three operational modes, including an Automatic mode with simpler operation when your objective is to create a maximally smooth or textured result while still preserving important details, an Advanced mode if you want more nuanced control, and an Ultra Beauty mode that provides additional controls that let you preserve fine detail that you want, while smoothing coarse details that you don't want.

Automatic

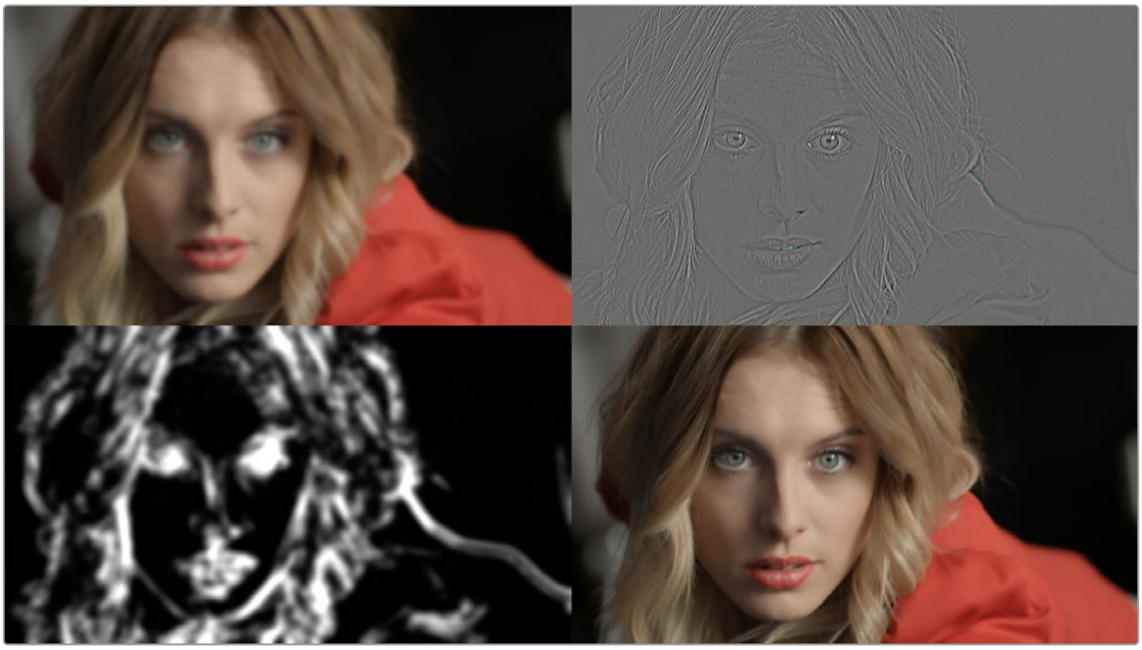
Automatic mode reveals easy-to-use controls for smoothing or coarsening detail.

- **Amount:** Lets you choose how much smoothing or coarsening to apply.
- **Scale:** Lets you reduce or increase the amount of smoothing or coarsening that's accomplished with the range of the Amount slider.

Advanced

Advanced mode reveals more manual controls for smoothing or coarsening detail, without the complexity of the Ultra Beauty mode.

- **Show Split View:** Checking this box puts a four up display in the Viewer showing the output of the Smoothing, Texture Recovery, Feature Recovery, and Final Output respectively. This is useful to have up when dialing in the advanced options below.



The Split View of the Advanced Beauty mode showing Smoothing (top left), Texture Recovery (top right), Feature Recovery (bottom left), and Final Output (bottom right) views.

Smoothing

These parameters let you remove imperfections. There are two smoothing controls to adjust. Apply these controls with enough strength to get rid of all the details to be smoothed away. The damage from over-smoothing will be reversed in Texture and Feature Recovery.

- **Smoothing Threshold:** Adjust this slider to set the amount of detail in the image to keep.
- **Diffuse Lighting:** Adjusts the amount of detail in the mid-tones.
- **Preview Smoothed:** Shows the result of just the smoothing operation in the Viewer.

Texture Recovery

These parameters let you restore areas of fine texture from the original video in flat areas away from the edges.

- **Texture Threshold:** Adjust this slider to set the amount of fine texture in the image to keep.
- **Add Texture:** Increases or decreases the scale of the texture in the Texture Threshold. Positive numbers add texture, while negative numbers remove it.
- **Preview Texture:** Shows the result of just the Texture Recovery operation in the Viewer.

Feature Recovery

These parameters let you determine how much of the recoverable Smoothing and Textures to bring back into the image.

- **Recovery Amount:** Adjust this slider to set the amount detail you've isolated (shown in white on the Feature Recovery view) to bring back into the image.
- **Preview Recovery Areas:** Shows the result of just the Feature Recovery operation in the Viewer.

Ultra Beauty

The Ultra Beauty controls reveal the full power of the Beauty plugin, with preview modes to examine different aspects of this plugin's behavior, while specifically adjusting the level of smoothing or coarsening you want, the amount of texture to preserve, and which additional features of the subject you want to recover from this operation to fine tune the result.

Smoothing

These parameters let you remove imperfections. There are two smoothing methods to chose from. Apply these controls with enough strength to get rid of all the details to be smoothed away. The damage from over-smoothing will be reversed in Detail Recovery. To see the effects of the smoothing alone, reduce the Detail Recovery strength, and texture to zero.

- **Flatten:** Reduces texture and filters out detail.

  Strength: How much smoothing effect is applied to the image.

  Levels: Controls how strongly abstracted the image is to smooth levels.

  Quality: Controls the sharpness of the resulting effect. Full being the sharpest and Fastest being the blurriest.

- **Filter:** Flattens smooth areas the image, leaving edges only where they divide flat regions from each other.

  Strength: How much smoothing effect is applied to the image.

  Filter Radius: Controls what scale of details are being filtered.

  Edge Threshold: Controls at what level an edge will be preserved or smoothed over. A low threshold will preserve even weak edges, while a high threshold will smooth over an increasing number of edges unless they are very high contrast.

Detail Recovery

These parameters let you restore the areas of the original video around the image edges, adding detail back into the image.

- **Strength:** Determines the amount of recovered detail.
- **Width:** Controls how far to either side of the edges image detail is recovered.

- **Gamma:** Controls whether to recover all edges, or only the stronger ones. A higher gamma reduces the amount of edges recovered.
- **Blur:** Softens the map of edges where detail is to be recovered.
- **Preview Edges/Details:** Enable this checkbox to show the areas where image detail is to be recovered; this allows you to see directly how the other controls in this section will affect the image.
- **Preview Recovery:** Enable this checkbox to show the actual detail being recovered alone over a neutral gray background.

Texture Recovery

These parameters let you restore areas of fine texture from the original video in flat areas away from the edges.

- **Texture:** Determines the amount of recovered texture to restore.
- **Scale:** Lets you set a threshold below which texture is extracted from the original image and applied to the result.
- **Detail/Edge Balance:** Lets you bias the recovery towards all textures (a negative value), or only to stronger details like edges (a positive value).
- **Preview Texture:** Enable this checkbox to show the actual texture being recovered alone over a neutral gray background.

Grain

These parameters let you add grain to the image to simulate detail in areas of the video with bad or no texture. Grain will only appear in smooth areas and will not cover the recovered edges.

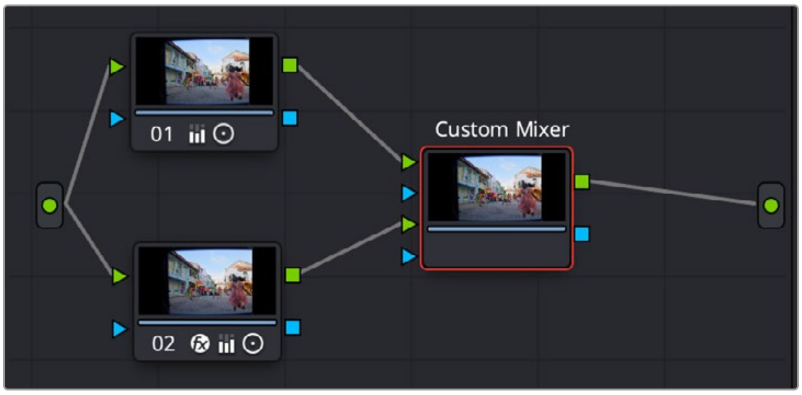
- **Strength:** Controls the amount of grain added to the image.
- **Size:** Controls the size of the individual grains.
- **Softness:** Softens the grain texture.
- **Saturation:** Applies saturation to the grain.

Custom Mixer

The Custom Mixer is an advanced version of the Layer node. It takes two inputs and combines them based on the Alpha channel of the second input. The Custom Mixer has controls to mix sources, blend effects, and interpolate between grades with per-channel control and mixing options. You can use this tool in a variety of ways to control the relative intensity between the sources, or simply as an alternative to the Layer Node but with slider controls.

The Custom Mixer should be added directly to the node graph and not dragged on top of a Corrector node. To add the Custom Mixer to your Node Graph, drag the Custom Mixer icon from the Resolve FX Refine section of the Effects library, directly into the Node Graph. You can also drag it on top of an existing link in the Node Graph to place it inline.

The Custom Mixer node is connected in the following way. The first and second RGB (Green) inputs are input 1 and input 2 respectively. The alpha of the second input controls the blend of input 2 onto input 1. The Alpha of the 1st input does not affect the result; it is passed through the node to the output key.



The Custom Mixer Node connections: Input 1 (Green triangle above), and Input 2 (Green triangle below). The alpha from Input 2 controls the blend of Input 2 onto Input 1 and is what you modify in the Custom Mixer controls.

The Custom Mixer has two main modes chosen by the Mix mode; both have the same behavior, just different controls to change how the input affects the output.

Blend

Blends the second input onto the first input. The normal range is from 0 (fully input 1) to 1 (fully input 2), however you can set the blend factor up to 10 times more or less by typing the numbers directly into the fields. This allows you to enhance the difference between inputs beyond the appearance of either one alone. All the sliders can be keyframed independently.

- **Blend Input 2 onto 1:** Controls the strength of each source, based on the color channels of the selected color space (RGB, YUV, XYZ, etc.). These can be adjusted separately if the Gang checkbox is unchecked.
- **Gang:** Check this box to gang the channels of the custom mixer together, or uncheck to adjust separately.

Combine

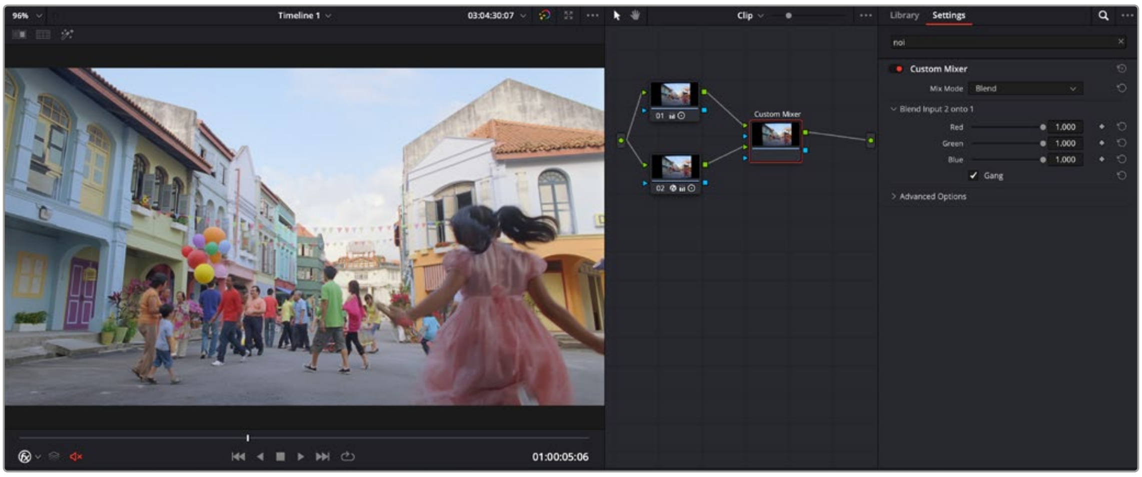
Combines the inputs together, allowing you to change the relative strength of each input's contribution to the mix. The normal range is from 0 (fully input 1) to 1 (fully input 2), however you can set the blend factor up to 10 times more or less by typing the numbers directly into the fields. This allows you to enhance the difference between inputs beyond the appearance of either one alone. All the sliders can be keyframed independently.

- **Input 1 Contribution:** Controls the strength of input 1's source, based on the color channels of the selected color space (RGB, YUV, XYZ, etc.). These can be adjusted separately if the Gang checkbox is unchecked.
- **Input 2 Contribution:** Controls the strength of input 2's source, based on the color channels of the selected color space (RGB, YUV, XYZ, etc.). These can be adjusted separately if the Gang checkbox is unchecked.
- **Offset:** Controls the offset of the combination, based on the color channels of the selected color space (RGB, YUV, XYZ, etc.). These can be adjusted separately if the Gang checkbox is unchecked.
- **Gang:** Check this box to gang the channels of the custom mixer together, or uncheck to adjust separately.

Advanced Options

Sets the composite type, as well as the color space and gamma of the resulting mix.

- **Composite Type:** Sets the composite type used for the mixer from the drop-down menu.
- **Color Space:** Sets the Color Space used for the blend. The default is the current Timeline color space.
- **Gamma:** Sets the Gamma used for the blend. The default is the current Timeline color space.



The initial setup of the Custom Mixer, Node 1 is the original image,
Node 2 is identical but with a Noise Reduction effect applied to it.

The final setup of the Custom Mixer, by extending the blend factor by 7,
the mixer actually adds noise to the image instead.

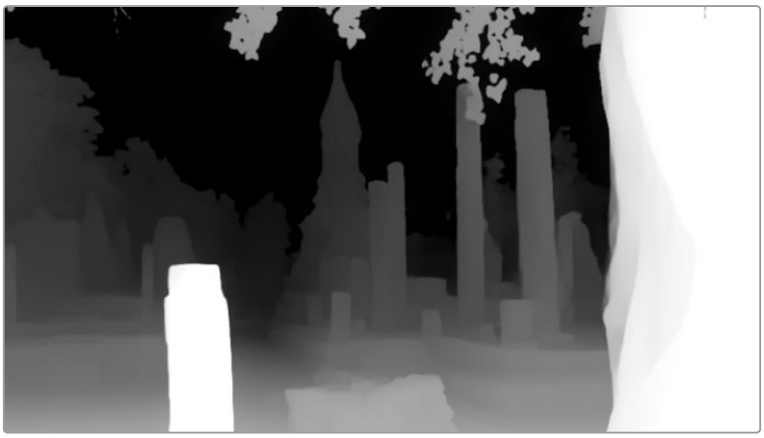
For the example above, by applying the Noise Reduction FX to Node 2, and directly connecting the same image from Node 1 to the Custom Mixer, the only difference between the inputs is the noise pattern from the effect itself. By manually turning the blend up past 1, you are blending in more of the noise than you started with. So by using Noise Reduction in conjunction with the Custom Mixer, you can actually add additional noise to an image that perfectly matches the original grain pattern without having to find a similar external grain or noise source.

Depth Map (Studio Version Only)

NOTE: The Depth Map Resolve FX has been considerably improved under the hood and updated from DaVinci Resolve 20 and above. While the interface and usage remains the same as the original, you should get significantly more precise results from the tool. The new Depth Map v2 has replaced the original and is applied automatically when using the Depth Map effect.

Depth Map creates an Alpha channel based on the perceived distance of objects in your clip. By being able to isolate a specific depth region, the opportunities to manipulate the resulting image are greatly expanded. For example, combined with a Fast-Noise plugin you could simulate a fog effect, but confined only to the distant background of your shot. You could enhance a foreground object like a person, increasing contrast, saturation, and sharpness, similar to using a qualifier. You can use a Depth Map to fix video issues, for example fixing a color temperature problem by isolating a far window that was shot in daylight causing background actors to be tinted blue, while leaving the foreground subject that was shot with studio lighting unaffected. Depth Map can also be combined with DaVinci Resolve's other windows and keying tools to form more complex rotoscoping and keys with less effort.

The resulting Depth Map Alpha channel is visualized as a black and white image, with white being the area that is affected by the resulting changes and black areas remaining unchanged.



(Top) The original image (Bottom) The Depth Map v2 analysis of the scene; white for areas "closer" to the camera and black for areas farther away

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Main Controls

These controls set the main parameters of the Depth Map.

- **Quality:** Depth Map is a very computationally intensive effect. The quality setting allows a Faster mode to speed up the responsiveness to adjustments, while the default Better mode gives the best results and should be turned on when the adjustments are finished.
- **Depth Map Preview:** By default this box is checked, and shows you the current Depth Map for making adjustments. When this check box is disabled, the resulting alpha can then be used for grading on other nodes. If you applied the effect to a corrector node, make sure to right click on the Depth Map's node and select "Use OFX Alpha" to have the node correctly pass the effect's Alpha channel.
- **Invert:** Checking this box reverses the Depth Map, switching its transparent and opaque regions.

Resulting Map Adjustment

These controls let you determine how the Depth Map's contrast is adjusted.

- **Adjust Map Levels:** When deselected (default), you can use DaVinci Resolve's grading tools (lift, gamma, gain, etc.) to adjust the full range of the Depth Map. When enabled, this option clips the Depth Map's levels to 0 and 1. This functions as a preview of what will happen to the Depth Map when used as an Alpha channel where the values are always clipped to 0 and 1 by DaVinci Resolve.
- **Far Limit:** This control adjusts the black levels of the Depth Map.
- **Near Limit:** This control adjusts the white levels of the Depth Map.
- **Gamma:** This control adjusts the intermediate depth values to be brighter or dimmer compared to the fixed black and white levels.

Isolate Specific Depth

These controls allow you to sweep backwards and forwards in the scene by depth, allowing you to isolate a specific depth range to adjust.

- **Isolation:** This checkbox turns the depth isolation tools on or off.
- **Target Depth:** This controls the specific depth you want to isolate. 1 is fully in the foreground, while 0 is fully in the background.
- **Tolerance:** Sets the range to either side of the Target Depth to include in the Depth Map.
- **Softness:** This sets a subtle ramp-in and ramp-out of the selected range, giving it a more organic selection.

Map Finesse

These controls modify the resulting Depth Map's Alpha channel for use in grading.

- **Post Processing:** This control turns the Map Finesse tools on or off.
- **Post Filter:** Smooths the depth map along image contours.
- **Contract/Expand:** This control dilates or erodes the overall shape at the edges, useful for fine tuning the boundary between the affected and unaffected regions of the map.
- **Blur:** This control softens the boundary of the map, allowing it to blend more smoothly into the resulting image.

Advanced Options

These controls tell the Depth Map how to handle various blanking issues.

- **Blanking Regions:** These controls tell the Depth Map how to handle letter and pillar boxing regions in the effect.

  Process Entire Frame: Tells the AI to ignore any information about the timeline shape and just process all the visible pixels.

  Handle Automatically (default): The AI will process only the area that DaVinci Resolve knows your media is in.

  Auto + Extra Crop: This is recommended for cropping away black bars; it uses DaVinci Resolve's knowledge of which pixels were occupied by your media but lets you crop out the framing or letterboxing relative to its shape. This is recommended because it will stay consistent as you adjust Timeline settings.

  Manual Cropping: Simply lets you override whatever the framing of the content is, ignoring any of DaVinci Resolve's knowledge of the shape of your media.

For example, say you have footage shot of a person on a rainy day, but the background is too distracting and contrasty. By creating a depth map that isolates only the foreground person, we can then apply a Fast Noise smoke effect to only the background, making it look like a foggy mist, and less distracting.

The original footage with a distracting background

The same footage with a Fast Noise Smoke effect node to simulate a fine mist, however it covers the entire frame, including the main subject. The mist belongs in the background only to be more realistic.

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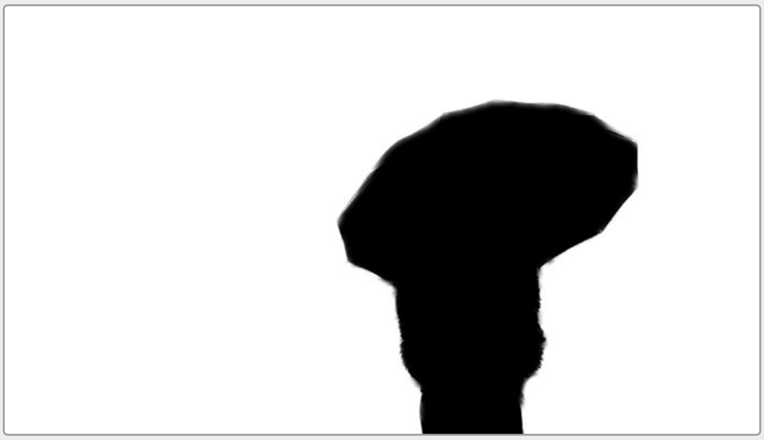


By placing a node with Depth Map applied between the original nodes, you can use the effect to pull a key based solely on "distance" from the camera. In this case, we have selected the background (white) and excluded the woman with the umbrella (black). Note the fine detail in the furry coat, which would be difficult to get with an HSL or window qualifier.

The final result of the process shows the main subject in the foreground with the Fast Noise smoke effect limited only to the background.

TIP: To create more believable Depth of Field adjustments, use the Depth Map effect in conjunction with the Tilt-Shift Blur effect. You can feed the results of the Depth Map directly into the Tilt-Shift Blur effect as a blur map by connecting the key output of the Depth Map node to the key input of the Tilt-Shift Blur node and selecting "From Alpha In" in the Map Source drop-down in the Depth of Field section of the effect. From there adjust the Focus Sweep to dial in the area you want to isolate. If you're not getting the expected results, try inverting the Depth Map.

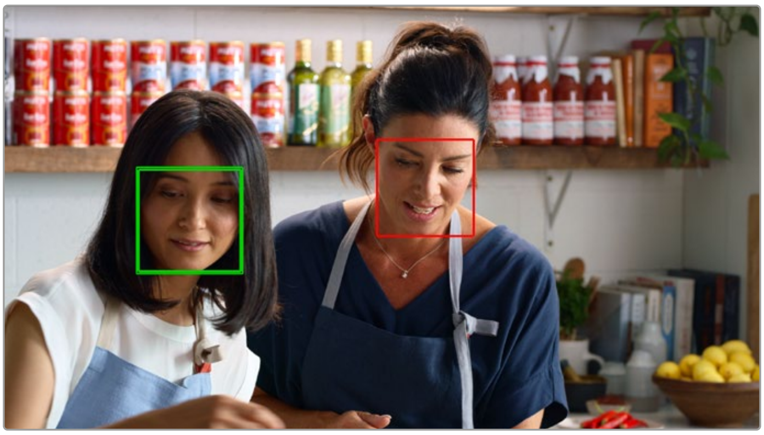
Face Refinement (Studio Version Only)

The Face Refinement effect has been improved with better tracking, profile handling, and smooth skin options. The newly revised effect is described below.

Face Refinement is an incredibly sophisticated, yet easy-to-use, filter that lets you quickly make very targeted adjustments to people's complexions.

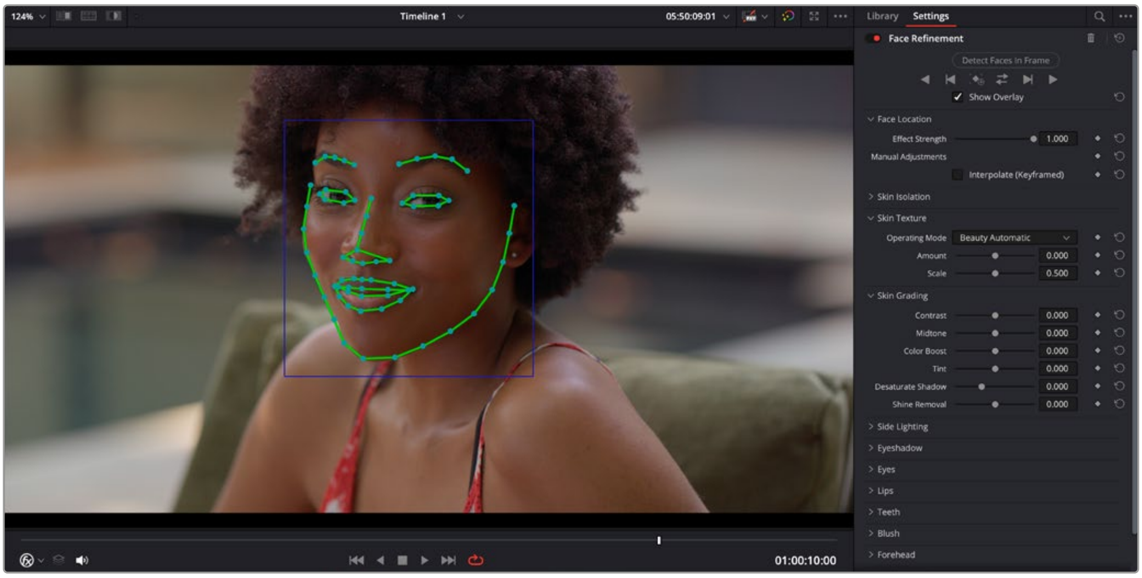
In order to use the face refinement toolset, the first thing you need to do is to identify the face (or faces) you wish to work on, and then analyze and track that face through the clip.

Identifying a face is done by clicking on the Detect Faces in Frame button. If faces are present in the frame at the position of the playhead, then this results in boxes being drawn over each detectable face. Click any of these boxes to choose which face you want to refine, and that box will be highlighted in green to indicate which face you'll be refining.



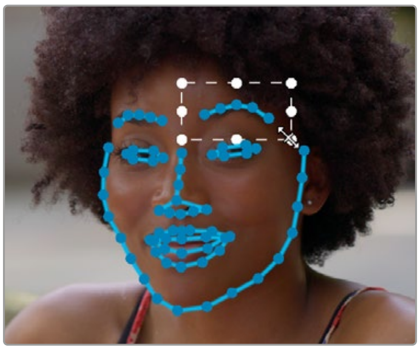
When multiple faces are detected, you can click a box to choose which one to work on.

After you've selected which face to work on, you need to use the tracking controls to track the face through the clip, and an outline of the face's major features appears to let you follow its progress.



The Face Refinement filter detecting a performer's face

The basic tracking controls usually work well enough to track faces in most instances. However, in some tricker cases if the subject's head tilts away, or if it's occluded by something else in the frame, you may have to manually adjust the track. You can do this by going to the last known good point of the track and then checking the Interpolate (Keyframed) box. This will add keyframes automatically for the frame and turn the outlines blue to let you know you are in manual mode. Then you can go to the problematic frames and click and drag to manually adjust the entire face or select and adjust each individual feature as needed. Once you get to a point where the automatic tracking works again, you simply uncheck the Interpolate box and let the automatic tracker take it from there.

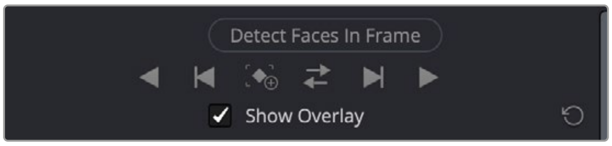


Manually adjusting an eyebrow in the Interpolate (Keyframed) mode

Once the face has been tracked, you're ready to work. This plugin divides the tracked face into different zones for common operations colorists often perform to quickly eliminate blemishes, adjust the hue and saturation of complexions in regionally appropriate ways, modify lighting, sharpen desirable detail, and refine makeup. Because all adjustments are fit to the proportions of the face that's been detected and tracked, all you need to do is make the adjustments you need and DaVinci Resolve takes care of the rest.

Main Controls

The top controls let you initiate the Face Refinement process.



The tracking controls for Face Refinement

- **Detect Faces in Frame:** This button initiates the process of using the facial detection of the Face Refinement plugin to detect the face you want to make adjustments for.

  Tracking Controls: Basic tracking controls to identify facial features motion through the clip

  Track Reverse: Identifies the face and tracks it backward through the clip.

  Track Reverse 1 Frame: Identifies the face and tracks it backward 1 frame.

  Find in Frame: Identifies the face in the current frame.

  Track Forward then Reverse: Identifies the face then tracks it forward from the current position and then reverse from that position.

  Track Forward 1 Frame: Identifies the face and tracks it forward 1 frame.

  Track Forward: Identifies the face and tracks it forward through the clip.

- **Show overlay:** To impress your clients and see how well Face Detection is tracking the face you're working on, you can turn on this checkbox, which turns on the wireframe tracking that shows you which facial details are being detected.

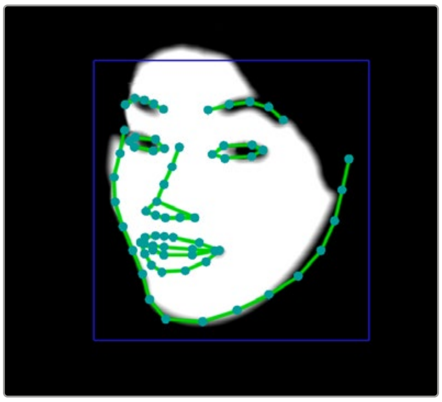
Face Location

These controls let you adjust the strength of the effect, as well as make manual adjustments to the face and its features position in the frame.

- **Effect Strength:** This slider determines the relative strength of the adjustments made with the original image. 1.000 is full strength, while 0 turns the effect completely off.
- **Interpolate (Keyframed):** Checking this box lets you make manual adjustments to the face and feature tracking in the Viewer, and automatically keyframes any changes. Click and drag in the Viewer to move the entire face. Shift-click and drag to reframe the face. Click on any individual feature to adjust, or any dot to move its position.

Skin Isolation

These controls let you adjust the skin mask this plugin automatically generates in order to limit the effect to only the face of the person you're targeting.



Adjusting the key with the face mask active

- **Show mask:** This checkbox makes the face mask being generated visible, which can be helpful to see while you're tuning it.
- **Skin Mask Source:** Lets you select the source for the skin mask. DaVinci Resolve's Internal Skin Model is the default, but there is the ability to use the Input Alpha if you've qualified the skin tones in another node. None disables the mask.
- **Shadow Range:** This control lets you extend or erode the shadows of the face detected by the skin model.
- **Tint Range:** This control lets you extend or erode the tint range detected by the skin model.
- **Temp Range:** This control lets you extend or erode the color temperature of the face detected by the skin model.
- **Limit to Face Region:** Turning on this checkbox places a circular garbage mask that follows the face to eliminate any accidental qualification of areas outside the face.
- **Region Size:** Lets you adjust the size of the face mask.
- **Region Softness:** Lets you adjust the edge softness of the face mask.
- **Flat Skin Areas Only:** Checking this box will exclude the nose and eye socket regions from the mask, allowing you to work only on the facial skin itself.
- **Exclude Mouth:** Checking this box will exclude the mouth region from the mask, allowing you to work only on the facial skin itself and not the lips.
- **Denoise mask:** This slider lets you fill in areas of small detail in the mask.
- **Refine mask:** This slider lets you adjust the face mask to smooth or eliminate gaps in the key.

Skin Texture

The Skin Texture controls have three operating modes that let you choose the method you want to use to control skin texture. Beauty Automatic and Beauty Advanced provide the texture controls available in the Beauty plugin, while Smoothing provides the previously available texture adjustment controls.

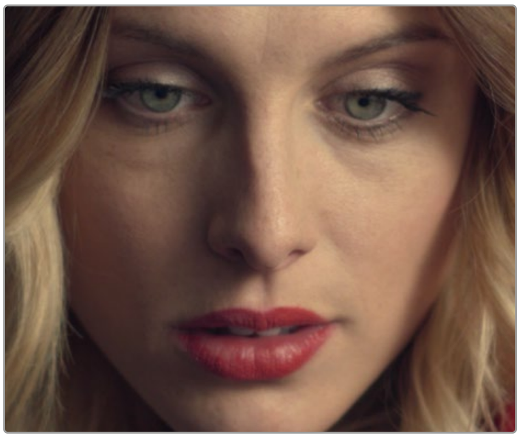
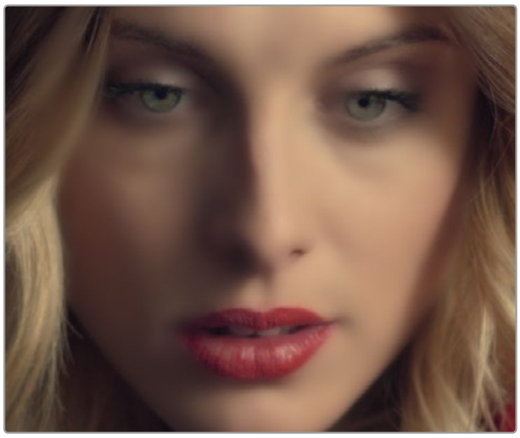
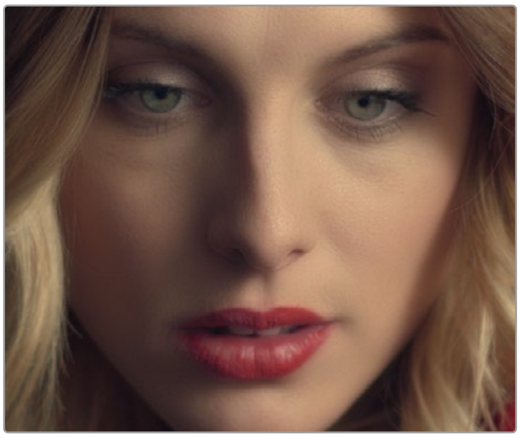
Beauty Automatic Controls

Automatic mode reveals easy-to-use controls for smoothing or coarsening detail.

- **Amount:** Lets you choose how much smoothing or coarsening to apply.
- **Scale:** Lets you reduce or increase the amount of smoothing or coarsening that's accomplished with the range of the Amount slider.

Smoothing Controls

- **Smoothing:** This slider removes detail from the areas isolated by the Skin Mask controls, smoothing the complexion. Its operation depends on the settings of the Detail Size and Detail sliders below. You may find you can increase Smoothing more dramatically as you increase Detail using the Detail Size and Detail sliders described below.
- **Detail Size and Detail:** Once you've used the Smoothing slider, you can then use these sliders to selectively put subtle skin details back into the image; Detail Size determines the maximum size of details you want to put back into the face, and Detail is a sharpening operation that lets you adjust how visible these small details are. By combining smoothing and subtle preservation of small details, you can get a much more naturalistic result than were you to simply leave the entire face unnaturally smooth.

(Top Left) The original image, (Top Right) Only smoothing complexion, (Bottom Left) Using Detail Size and Detail to put natural texture back into the smoothed result (results exaggerated for print)

Beauty Advanced

These controls are identical to the Advanced controls of the Beauty plugin, covered previously in this chapter.

Color Grading

These controls let you make color adjustments to the overall face.

- **Contrast:** This slider lets you lighten the face in a natural way by keeping the shadows dark, even while you brighten the face, making it easy to bring actors out of the background.
- **Midtone:** Lets you add a more luminous quality to the skin tone.
- **Color boost:** Specifically boosts saturation in the lowest saturated parts of the face.
- **Tint:** Provides a limited range of naturalistic hues emphasizing orange through red (but extending to green and magenta) with which to tweak complexion.
- **Desaturate shadow:** This slider lets you selectively desaturate the darkest shadows on the face to keep things looking natural, or to desaturate more aggressively in order to stylize the face in a different way. Also useful when using the other adjustments of this plugin makes the shadows too colorful. Moving this slider into negative values will add saturation.
- **Shine Removal:** This slider is an inverse contrast adjustment designed to ameliorate sweat and shine on a person's face, although moving this slider into negative values will also accentuate shine.

Side Lighting

These controls allow you to fill in or accentuate shadows and balance the light between the sides of the face.

- **Lift Left/Right:** Adjusts the dark parts of the image for a side of the face.
- **Gamma Left/Right:** Adjusts the midtone parts of the image for a side of the face.
- **Gain Left/Right:** Adjusts the bright parts of the image for a side of the face.
- **Area Size:** This control lets you adjust the size of the area of influence of the side lighting.
- **Bias Left/Right:** Normally the left and right sides of the face are split right down the nose. You can use this control to manually adjust the center of the effect, if necessary.
- **Area Softness:** This control adjusts the softness of the effect around the side of the face.

Eye Shadow

These controls give extensive control of the eyelids and sockets, without manipulating the eyes themselves. Used subtly, it can easily remove bags under the eyes, or accentuate makeup details. Used aggressively, you can create realistic eye makeup effects from nothing. There are separate controls for the upper and lower eyelid areas.

- **Temp Upper/Lower:** Adjusts the color temperature of the eyelid.
- **Tint Upper/Lower:** Adjusts the tint of the eyelid.
- **Hue Upper/Lower:** Adjusts the color of the eyelid.
- **Saturation Upper/Lower:** Adjusts the saturation of the eyelid.
- **Gamma Upper/Lower:** Adjusts the gamma of the eyelid.
- **Size Upper/Lower:** Adjusts the size of the area of the eyelid modifications.
- **Softness Upper/Lower:** Adjusts how soft the blending is between the eyelids and the face.
- **Strength Left/Right:** Lets you set different strengths for the left or right eye to compensate for lighting differences between the sides of the face.


(Left) The original image, (Right) using the Eyeshadow tools to change the model's upper eyeshadow color to purple and increase the intensity

Eye

These controls target just the eyes and surrounding area of the face.

- **Sharpening:** This slider lets you selectively add sharpening just to the eyes and eyelashes, which lets you instantly add focus to any performer.
- **Brightening:** This slider lets you whiten the eyeballs.
- **Eye Light:** This slider lets you brighten the region of the face around the eyes that's often thrown into shadow by the subject's forehead in situations with imprudent lighting.
- **Eyebag Removal:** This slider uses a variety of techniques to smooth, color-adjust, and brighten the area under the eyes most susceptible to eye bags with tired talent.


(Left) Eyes with no special adjustment, (Right) eyes with adjustment

Lips

These controls target just the lips and surrounding area of the mouth.

- **Hue Shift:** This slider lets you adjust the color of a subject's lips or lipstick.
- **Saturation:** This slider lets you adjust the intensity of lip color.
- **Gamma:** This slider lets you adjust the lip brightness using a gamma curve.
- **Upper Lip Smooth:** Lets you smooth out fine age lines that can appear above lips.
- **Mouth Corners Inner/Outer:** Adjusts the side corners of the mouth.
- **Top Inner/Outer:** Adjusts the top boundary of the lips.
- **Bottom Inner/Outer:** Adjusts the bottom boundary of the lips.
- **Softness:** The softness of the blend between the mouth area and the rest of the face.

Teeth

A smile can hide a thousand lies, but just these three controls can accentuate the teeth.

- **Blue Gain:** This control is used to color balance the teeth, making them whiter or yellower as desired.
- **Gamma:** This manipulates a gamma curve allowing you to set how bright a smile the subject has.
- **Mask Softness:** The softness of the blend between the teeth and the rest of the face.

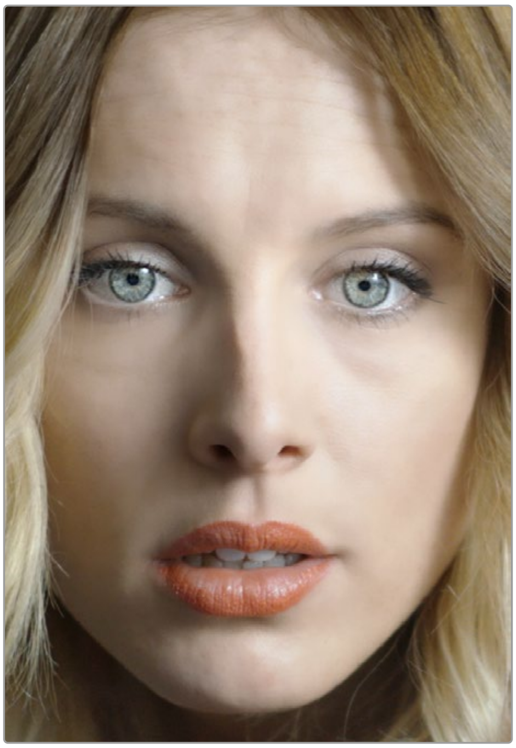
Blush

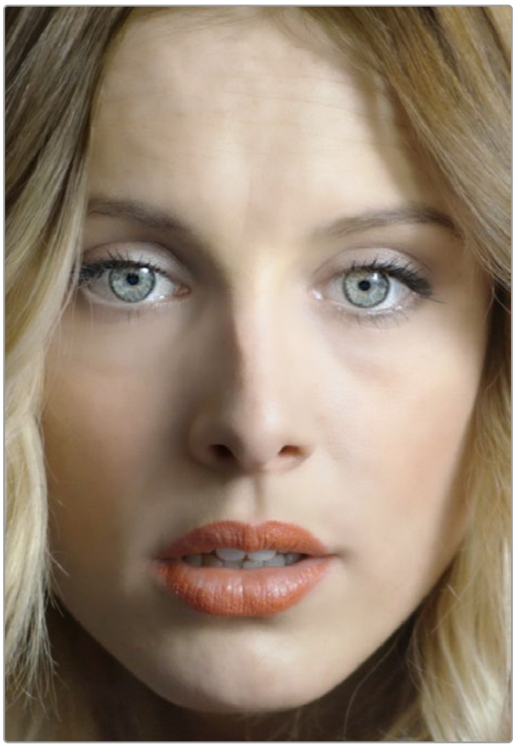
The parameters in this group let you modify the hue of the blush region of a subject's cheek, letting you correct an unwise makeup choice, or push a subject's blush color around to add makeup that wasn't there.

- **Hue:** This slider lets you adjust the hue of the cheeks.
- **Saturation:** This slider then lets you intensify or remove blush color.
- **Size:** This slider lets you adjust the size of the blush area of the cheeks.

About Forehead, Cheeks, and Chin Retouching

The next three groups of controls borrow a technique from portrait painters, who've long taken advantage of the "traffic light" approach to rendering skin hues that observes that foreheads are often a bit yellow, the middle of the face is usually a bit red, and chins can be a bit green. A combination of unequal sun exposure, capillary distribution, and follicle growth account for this, but the bottom line is that faces are seldom a single unified hue. This means that there may be a region of the face that would benefit from individual adjustment (cheeks that have gotten too much sun, for example). But it also means that when you allow a bit of hue variation into your face grade, you can achieve a more naturalistic result.





(Left) A face graded with a single hue, (Right) a face graded with slight variation in the forehead, blush area, and chin

TIP: For a variety of reasons, people are extremely sensitive to the hues of skin tone, so tiny variations that can be difficult to identify nevertheless have a significant impact on the resulting visuals. Unless you're aiming to create a special effect, these controls are meant to be used sparingly.

Forehead

As the name implies, adjusts color and texture on the forehead.

- **Hue and Saturation:** These sliders let you adjust forehead color.
- **Gamma:** This manipulates a gamma curve allowing you to set how bright the forehead is.
- **Smooth:** This slider lets you apply a specific smoothing operation to the forehead to ameliorate wrinkles and worry-lines.

Cheeks

Simple color adjustment controls affecting the entire cheek area, and not just the blush area.

Hue and Saturation: These sliders let you adjust the color of the cheek, eye, and nose area.

Chin

Simple color adjustment controls over the chin of the face.

Hue and Saturation: These sliders let you adjust the color of the chin, running up alongside the sides of the face (the beard area).

Relight (Studio Version Only)

Relight allows you to add light sources to your scene, realistically illuminating the objects within it. Unlike a Power Window whose shape is fixed on screen, Relight analyzes the shapes of people and objects to create a Surface Map and uses that map to reflect light appropriately. When dialed-in properly, this effect can provide the illusion of light following the curves of an object, reflecting off highlights, and disappearing into shadows.

Relight is useful in situations when you wished you had another light in the scene, want to accentuate an existing light to match another shot, or increase the realism of day-for-night grading.

NOTE: The Relight effect creates the illusion of depth with its analysis of the surfaces in a scene, but it does not calculate 3D relationships between objects in the scene. Shapes cannot be made to cast shadows, and depth information cannot be produced.

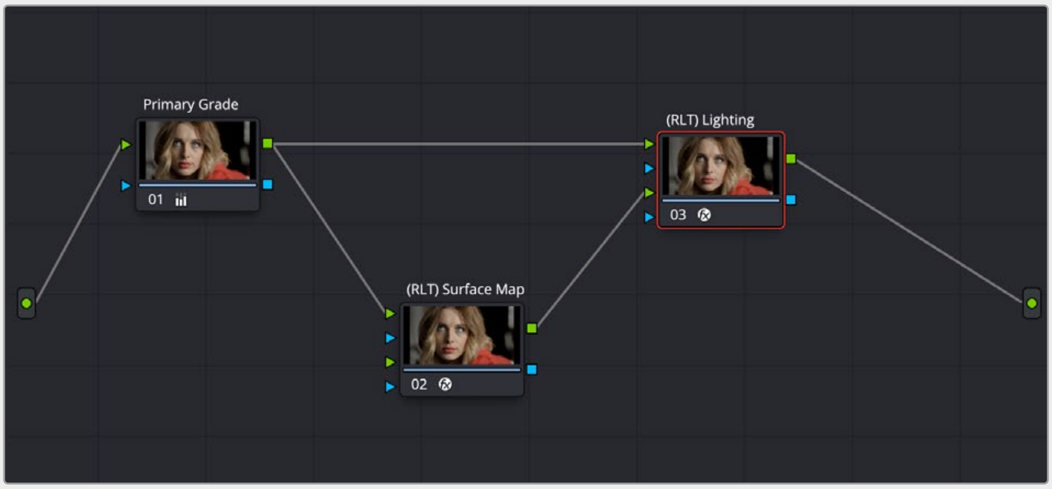
Setting up Relight

Analyzing the image to create a Surface Map is very computationally intensive. Calculating the effects of light sources to an existing analysis is comparatively very fast. By default, adding Relight to a node will create the map and add a light in one step. This is convenient, but adjusting the light position re-runs the effect, and hence re-runs the analysis. So it can be slow to work with unless you have an extremely powerful computer. Fortunately it is possible to split the work across two nodes, caching the analysis in the first node and adjusting the light positions in the second node. This is described in detail below.

Note that you can add multiple light sources reusing the same analysis. If you have, say, three separate lights in the same scene, reusing one analysis, then this will run three times faster than repeating the analysis for each light.

And as an advanced option, the split approach allows you to use a Surface Map which has been saved out to file, or a Surface Map from other software. Computer graphics software can often render "surface normals" of 3D models along with the images they create and these can be made compatible with Relight. In this context a "normal" is a geometric term describing which way a surface is oriented. There is no universal format for surface normals; see the External Map settings for some options which can help with using these as Surface Maps in the plugin.

Here is a sample node graph splitting the Relight in two:



How to set up a dual Relight Node Tree with two separate Relight nodes, configured as above.
These have been labeled Surface Map and Lighting for clarity.

In the above example, you have the source video and two Relight nodes. This is how they're set up:

- The source video is connected to the first RGB input (top green triangle) of both nodes.
- The first Relight Node ((RLT) Surface Map) creates the Surface Map for the image by selecting "Output Surface Map" checkbox in the Relight effect's settings. When checked, the plugin goes into analysis-only mode and other options disappear. You can enable Node Caching to automatically generate the Surface Map in the background when you aren't using DaVinci Resolve. The RGB output of this node is connected to the second RGB input (bottom green triangle) of the second Relight Node.
- In the second Relight Node ((RLT) Lighting), set the first option "Surface Map" from its default ("Use Internal") to "Use Input 2." The other settings all remain visible; this node is where you will adjust the light properties.

You can add Power Windows, Qualifiers, or masks to either node, or to separate nodes and connect them via the first Alpha channel (top blue triangle) of either node. For example, you could select a person using the magic mask and use it to limit the lighting to just that person and not the background.

The output of the Relight Surface Map.
This goes into the input of the Relight Lighting node.

Using Relight

When first placing Relight on a node and turning off the lighting preview, there is no visible change to the image. This is because the map of what the light is illuminating is in the Alpha channel, ready for all of DaVinci Resolve's color grading tools to go to work. The simplest place to start is to increase Gain (dial it up directly or add color); often changes to gamma and saturation may be appropriate.

For a day-for-night effect you can invert the Alpha to darken all of a scene except the light areas, to achieve a day-for-night effect but with a "light source" that is the original scene lighting.

The Relight Settings

These controls allow you to adjust the intensity, type, and direction of the light that reflects off the surface map. Most are keyframable, and positions can be attached to the FX Tracker, allowing things like moving a light source as if it were attached to an object, or moving a spotlight to follow a person as they move through the scene. Controls and their groups are:

- **Surface Map:** This chooses which mode you will use to create the Surface Map.

  Use Internal: Generates the Surface Map and immediately uses it in this node.

  Use Input 2: Receives the Surface Map from another Relight Node. Use this on the second node of a dual Relight Node setup.

- **Output Surface Map:** Checking this box causes this copy of the effect to output a multi-colored map of the input frame. This Surface Map can then be fed into a second Relight effect (set to "Use Input 2" as above) where the light sources are placed and adjusted.
- **Directional/Point Source/Spotlight:**

  Directional: Creates a light source that emits from a specific direction at an infinite distance.

  Point Source: Creates a light source that emits from a central point.

  Spotlight: Is also a point source but it is directed into a cone shape.

COLOR

- **Relighting Map Preview:** Checking this box reveals the Alpha channel of the effect of this light. Unchecking this box shows the current image but with the light mask in the Alpha channel.
- **Diffuse Reflection Properties:** These controls determine the quality of the emitted light.

  Brightness: This controls the strength of any grading that is applied to the scene. It does not add brightness changes on its own but amplifies the changes you make in the grading tools.

  Softness: This controls the size of the light emitter, effectively determining how harsh or gentle the gradient is between light and dark.

  Distance Falloff: This controls how much reflections will dim with distance to the light source.

- **Light Properties:**

  Brightness: This controls the strength of any grading that is applied to the scene. It does not add brightness changes on its own but amplifies the changes you make in the grading tools.

  Reach: This controls how much reflections will dim with distance from the light source.

  Contrast: This determines how harsh or gentle the gradient is between light and dark.

- **Surface Properties:**

  Glossiness: Gives the illusion of a metallic sheen to reflective surfaces.

  Specularity: When there is any glossiness added, this controls how metallic/shiny that glossiness is.

  Shadow Softness: Controls how much the light penetrates into shadows due to ambient reflections, beyond where it would reach otherwise.

- **Spotlight :**(shown only with Spotlight option enabled)

  Beam Angle: Sets how broad the cone of light is. This control is also available graphically in the Viewer.

  Edge Softness: Controls the softness of the edges of the light cone/beam.

- **Light Position:** These controls are for the the position of the light emitter for all light types. They can be adjusted directly, but more easily, they can be adjusted graphically in the Main Viewer. Ensure that the Main Viewer's mode is set to Open FX Overlay with the icon at the bottom left.

  In Directional mode:

  Azimuth/Elevation XY : Controls the direction the light is coming from.

  In Point and Spotlight modes:

  Source Follows FX Tracker: Check this box to have the light source follow the FX tracker.

  Light Source XYZ: These controls determine the light emitter's position in 3D space.

  In Spotlight mode:

  Spot Follows FX Tracker: Check this box to have the light source follow the FX tracker.

  Spot Target XY: These controls determine the position of the base of the spotlight cone.

When you are in Use Input 2 mode, there are options for using an externally-generated Surface Map.

- **Rescale Oversaturated:** Different applications use different scalings of the Surface Map. A common format looks like the Relight plugin's internal ones but with the colors intensely saturated and can be dark or black in some areas. If your imported surface map has this appearance, check this box to re-interpret it.
- **Reinterpret Left/Right:** Check this box to swap the direction of your external surface map if your map surfaces are appearing concave instead of convex.
- **Reinterpret Up/Down:** Check this box to swap the direction of your external surface map if your map surfaces are appearing lit from below when the light is above them.

When you are generating an analysis internally with Use Internal, or you are using Output Surface Map, you will also see an Advanced Options group with an option related to Blanking Regions. These will not apply in the majority of cases; they are only for fixing some specific situations. They are for forcing different behavior at the edges of the frame in these special cases:

- The source video has letterboxing or some framing border "baked in," meaning that the incoming frames have black pixels at the edges that were not added by DaVinci Resolve. Letterboxing added by DaVinci Resolve (e.g., a 21:9 frame fit to a 16:9 timeline) does not need any special treatment. You will recognize that there is a problem because the Surface Map AI will treat the scene as if the black areas are actually objects or attached to objects in frame. Objects in contact with the black bars may appear to blend into those areas.
- The source video has pillarboxing, same as with letterboxing above. For example, if a 4:3 source video was captured to a 16:9 clip with black borders before it was used in DaVinci Resolve.
- A case where a wide clip was inserted into a narrower timeline correctly, but after resizing or effects were applied you wish for the plugin to process the original blanking areas because there is now content there. This case will appear obvious, in that the analysis will stop at the original blanking boundary, cropping-away areas you want to analyze.

If you encounter any of these special cases your options are:

- **Blanking Regions:**

  Process Entire Frame: Tell the AI to ignore any information about the timeline shape and just process all the visible pixels.

  Handle Automatically (default): The AI will process only the area that DaVinci Resolve knows your media is in

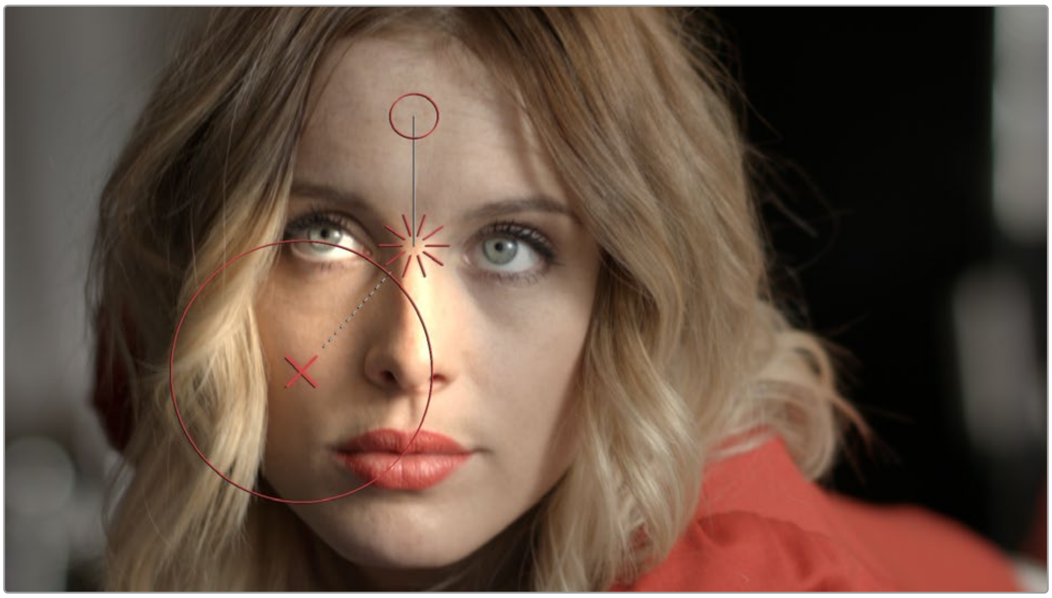
  Auto + Extra Crop: This is recommended for cropping away black bars, it uses DaVinci Resolve's knowledge of which pixels were occupied by your media but lets you crop out the framing or letterboxing relative to its shape. This is recommended because it will stay consistent as you adjust timeline settings.

  Manual Cropping: Simply lets you override whatever the framing of the content is, ignoring any of DaVinci Resolve's knowledge of the shape of your media.

The Relight On-Screen Controls

Several of the controls for positioning the light source are available graphically in the Viewer; make sure that Open FX Overlay is selected in the lower left of the Viewer to see them.

In the Directional mode, you can drag the emitter (red rays) around the frame to set the Azimuth/Elevation controls.

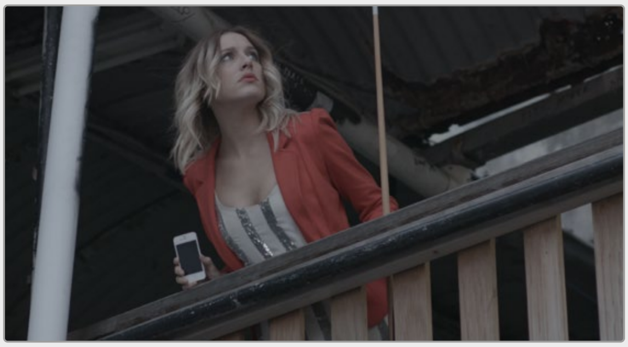


In the Point Source and Spotlight modes, you can move the emitter (red rays) to change XY coordinates, and the circle to adjust the emitter's Z coordinates. If in Spotlight mode you can drag the X to change the XY coordinates of the Spot Target, and scale the large circle to adjust the Cone Angle.

COLOR

Example

In this example, we will be relighting the scene below. Shot during a cloudy overcast day, the director likes the flat look on the building. Unfortunately, the muted drab lighting also affected the model, which he wants to pop more out of the scene. We are going to use the Relight tool to add a phantom sunbeam directly to the model and not to any of the surrounding fence or walls.

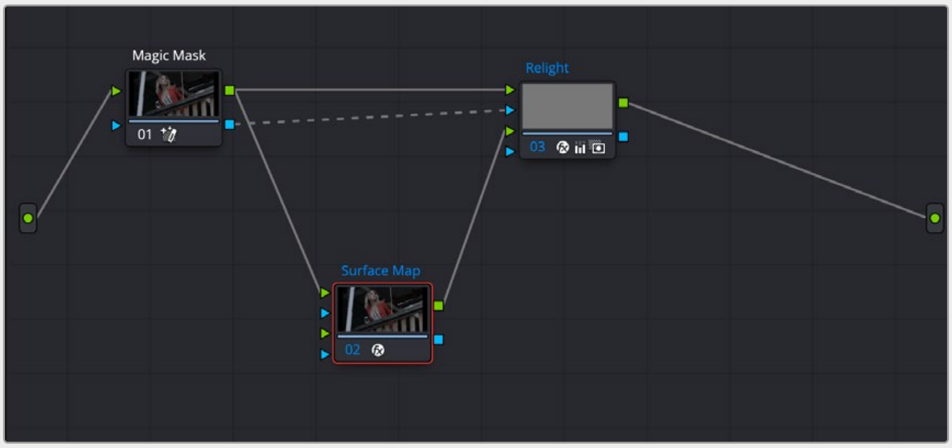


The model's lighting is flat and drab. We want to change it by adding an artificial sunbeam falling on her that wasn't there in the shot. We also want to keep the fence and background as they are.

The first tool we will use is the Magic Mask to isolate the model from her environment.

Isolating the model with a Magic Mask to isolate the effect

Next we will add two different relight nodes, one to create the surface map, the other to receive it and actually relight the shot. The node tree is explained below.



The Relight node tree; there are two Relight nodes, one called Surface Map and one called Relight.

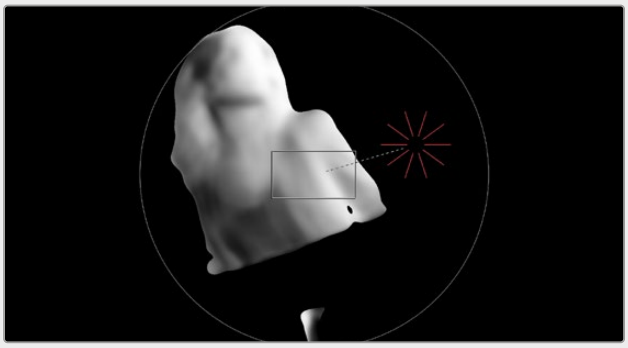
COLOR

We feed the RGB output of the Magic Mask to the top RGB inputs of both nodes. We also feed the Alpha output of the Magic Mask into the Relight node. This way the Relight effect will only apply to the model we selected using the Magic Mask and leave the background alone.

In the first node (Surface Map), we select Use Internal from the Surface Map option. We also check the Output Surface Map checkbox to pass that generated map to the second (Relight) node. We then complete the connection by dragging the RGB output of the Surface node to the bottom RGB input of the Relight node. In the Relight node, we then select Use Input 2 in the Surface Map option in the Inspector.

The surface map generated by the Surface Map node

Now that we have the node tree setup correctly, the Surface Map is generated in its node; make sure it's node cache is set to On. Then we can work directly in the Relight node and select a Directional light. Using the Relighting Map Preview checkbox and the on screen controls, we can then see exactly where the lighting effect will take place, and how it will follow the curves of the subject.



With the Relighting Map Preview box checked, you can see exactly where and how the Relight will affect the subject. Unchecking the box lets you see the final color output.

The last thing we need to do is actually create our new light properties, so by using the Primaries Wheels controls in the Relight effect, we can increase the gain and color it slightly yellow/orange and increase the saturation to mimic a golden hour sunbeam on our subject, where there was only a flat cloudy day before.

Adjusting the Primaries gain and saturation wheels in the Relight node to add the character of the light, selling the illusion that the model is leaning over the fence into a sunbeam.